

A Log-Free p -Triangle Inequality for $L^{p,r}$

A candidate full solution to an open problem in arXiv:1801.10570

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Status. Candidate full solution, likely valid, awaiting expert review. The theorem removes the logarithmic factor whose necessity is asked about by Seeger–Trebls. The proof is self-contained and quantitative for the Lorentz quasi-norm normalization used in the source.

1 Source problem

Let (X, μ) be a nonatomic measure space. For $0 < p, r < \infty$, write

$$\|f\|_{p,r} = \left(\frac{r}{p} \int_0^\infty [t^{1/p} f^*(t)]^r \frac{dt}{t} \right)^{1/r}, \quad \|f\|_{p,\infty} = \sup_{t>0} t^{1/p} f^*(t). \quad (1)$$

Here f^* is the nonincreasing rearrangement of $|f|$.

Seeger and Trebls prove that, when $0 < p < 1$ and $p < r < \infty$,

$$\left\| \sum_k f_k \right\|_{p,r} \leq C(p,r) \left(\sum_k \|f_k\|_{p,r}^p \right)^{1/p}. \quad (2)$$

Their bound for $C(p,r)$ contains

$$\left(1 + \frac{p}{r} \log \frac{1}{1-p} \right)^{1/p-1/r}.$$

They ask explicitly whether this logarithmic term is necessary [1, Open problem after (13), and Appendix B]. We give a negative answer.

2 Idea of the proof

Put $\theta = 1 - p/r$, so that $r = p/(1 - \theta)$. The two endpoints behind the result are transparent. At $r = p$, inequality (2) is the pointwise p -triangle inequality in L^p and has constant one. At $r = \infty$, a common amplitude cutoff gives a weak- L^p estimate of size $(1 - p)^{-1/p}$.

The main point is to interpolate these endpoints without invoking an abstract quasi-Banach interpolation theorem or losing control of constants. For every simple f , nonatomicity supplies a rank coordinate ρ with $|f| = f^*(\rho)$. The exact factorization

$$|f| = u^{1-\theta} v^\theta, \quad v = \rho^{-1/p}, \quad u = |f|^{1/(1-\theta)} \rho^{\theta/[p(1-\theta)]}$$

places u in L^p and v in weak L^p . Its normalization is explicit:

$$\|u\|_p^p = (1 - \theta)\|f\|_{p,r}^r, \quad \|v\|_{p,\infty} \leq 1.$$

After rescaling each pair of factors, discrete Hölder turns a sum of products into a product of sums. A two-threshold rearrangement inequality then returns that product to $L^{p,r}$. Optimizing the two thresholds produces only the binary-entropy factor $\theta^{-\theta}(1 - \theta)^{-(1-\theta)} \leq 2$, not a logarithm.

3 The log-free theorem

For $0 < \theta < 1$, set

$$\mathcal{H}(\theta) = \theta^{-\theta}(1 - \theta)^{-(1-\theta)}. \quad (3)$$

As usual, the endpoint values are interpreted by continuity, and $1 \leq \mathcal{H}(\theta) \leq 2$.

Theorem 1 (Entropy-refined log-free bound). *Let $0 < p < 1$ and $p < r < \infty$, and put $\theta = 1 - p/r$. On every nonatomic measure space, every finite or countable family with $\sum_k \|f_k\|_{p,r}^p < \infty$ satisfies*

$$\left\| \sum_k f_k \right\|_{p,r} \leq \mathcal{H}(\theta)^{1/p} p^{-\theta} (1 - p)^{-\theta/p} \left(\sum_k \|f_k\|_{p,r}^p \right)^{1/p}. \quad (4)$$

Consequently, with an absolute constant $A = 2e^{1/e}$,

$$C(p, r) \leq A^{1/p} \left(\frac{1}{1 - p} \right)^{1/p - 1/r}. \quad (5)$$

In particular, the logarithmic factor in the Seeger–Trebel’s bound is not necessary.

The rest of the packet proves the theorem from three elementary lemmas.

4 Three lemmas

4.1 A self-contained weak endpoint

Lemma 2 (Weak p -summation). *For $0 < p < 1$ and every finite family of measurable functions,*

$$\left\| \sum_k |g_k| \right\|_{p,\infty} \leq D_p \left(\sum_k \|g_k\|_{p,\infty}^p \right)^{1/p}, \quad D_p = p^{-1}(1 - p)^{-1/p}. \quad (6)$$

Proof. Put $a_k = \|g_k\|_{p,\infty}$ and $S^p = \sum_k a_k^p$. Fix $\alpha > 0$ and a common cutoff $\beta > 0$. Let

$$E_\beta = \bigcup_k \{|g_k| > \beta\}, \quad h_k = |g_k| \mathbf{1}_{\{|g_k| \leq \beta\}}.$$

The definition of the weak norm gives

$$\mu(E_\beta) \leq \beta^{-p} S^p. \quad (7)$$

Layer cake and the same weak estimate give

$$\int_X h_k d\mu \leq \int_0^\beta \mu(|g_k| > s) ds \leq \frac{a_k^p \beta^{1-p}}{1 - p}. \quad (8)$$

Outside E_β , the sums of the $|g_k|$ and the h_k agree. Hence Chebyshev's inequality, (7), and (8) imply

$$\mu\left(\sum_k |g_k| > \alpha\right) \leq S^p \left(\beta^{-p} + \frac{\beta^{1-p}}{(1-p)\alpha}\right).$$

The choice $\beta = p\alpha$ minimizes the displayed expression and makes it

$$S^p \alpha^{-p} \frac{p^{-p}}{1-p}.$$

Taking the supremum of $\alpha \mu(\sum_k |g_k| > \alpha)^{1/p}$ proves (6). $\square$

4.2 Quantitative rank factorization

Lemma 3 (Strong-weak factorization). *Let $0 < p < 1$, $0 < \theta < 1$, and $r = p/(1 - \theta)$. If f is a nonnegative simple function with support of finite measure on a nonatomic measure space, then there are $u, v \geq 0$ such that*

$$f = u^{1-\theta} v^\theta, \quad \|v\|_{p,\infty} \leq 1, \quad \|u\|_p^p = (1 - \theta) \|f\|_{p,r}^r. \quad (9)$$

Proof. Write the positive values of f in decreasing order as $c_1 > \dots > c_m > 0$ and let $E_j = \{f = c_j\}$. Set $M_j = \sum_{i \leq j} \mu(E_i)$ and $M_0 = 0$. Nonatomicity, by recursive bisection, supplies on every E_j a measurable variable uniformly distributed on (M_{j-1}, M_j) . Combining these variables gives a rank map

$$\rho : \text{supp } f \longrightarrow (0, M_m)$$

whose pushforward is Lebesgue measure and for which $f = f^*(\rho)$ almost everywhere. This also handles flat levels: each level is spread across its full rearrangement interval.

Define

$$v = \rho^{-1/p}, \quad u = f^{1/(1-\theta)} \rho^{\theta/[p(1-\theta)]}.$$

The factorization in (9) is immediate. Since ρ is uniform,

$$\mu(v > \lambda) = \min\{M_m, \lambda^{-p}\},$$

and therefore $\|v\|_{p,\infty} \leq 1$. Also, using $r = p/(1 - \theta)$ and the normalization (1),

$$\begin{aligned} \|u\|_p^p &= \int_0^{M_m} f^*(t)^{p/(1-\theta)} t^{\theta/(1-\theta)} dt \\ &= \int_0^\infty f^*(t)^r t^{r/p-1} dt = (1 - \theta) \|f\|_{p,r}^r. \end{aligned}$$

This proves all assertions. $\square$

4.3 Returning a product to the Lorentz space

Lemma 4 (Product rearrangement). *Let $0 < p < \infty$, $0 < \theta < 1$, and $r = p/(1 - \theta)$. Then*

$$\|u^{1-\theta} v^\theta\|_{p,r} \leq K_\theta \|u\|_p^{1-\theta} \|v\|_{p,\infty}^\theta, \quad (10)$$

where

$$K_\theta = (1 - \theta)^{-2/r} \theta^{-\theta/p}. \quad (11)$$

Proof. For every $0 < a < 1$ and $t > 0$, the elementary two-threshold rearrangement bound is

$$(u^{1-\theta}v^\theta)^*(t) \leq u^*(at)^{1-\theta}v^*((1-a)t)^\theta. \quad (12)$$

Indeed, away from the union of the sets where u and v exceed the two right-hand thresholds, whose total measure is at most t , the product is at most the right-hand side. The weak estimate for v and $r(1-\theta) = p$ now give

$$\begin{aligned} \|u^{1-\theta}v^\theta\|_{p,r}^r &\leq \frac{r}{p}(1-a)^{-\theta r/p} \|v\|_{p,\infty}^{\theta r} \int_0^\infty u^*(at)^p dt \\ &= \frac{r}{pa}(1-a)^{-\theta r/p} \|u\|_p^p \|v\|_{p,\infty}^{\theta r}. \end{aligned}$$

Thus (10) holds with

$$\left(\frac{r}{pa}\right)^{1/r} (1-a)^{-\theta/p}.$$

Choosing $a = 1 - \theta = p/r$ yields (11). $\square$

5 Proof of Theorem 1

We first take finitely many nonnegative simple functions of finite-measure support. Put

$$F_k = \|f_k\|_{p,r}, \quad S = \left(\sum_k F_k^p\right)^{1/p}.$$

Ignore zero terms. Lemma 3 gives $f_k = u_{k,0}^{1-\theta}v_{k,0}^\theta$ and

$$\|u_{k,0}\|_p^{1-\theta} \|v_{k,0}\|_{p,\infty}^\theta \leq (1-\theta)^{1/r} F_k. \quad (13)$$

Rescale the two factors reciprocally, preserving their product, so that the new factors u_k, v_k obey

$$f_k = u_k^{1-\theta}v_k^\theta, \quad \|u_k\|_p = F_k, \quad \|v_k\|_{p,\infty} \leq (1-\theta)^{1/(r\theta)} F_k. \quad (14)$$

For completeness, if $u_k = c_k u_{k,0}$ with $c_k = F_k/\|u_{k,0}\|_p$, take $v_k = c_k^{-(1-\theta)/\theta} v_{k,0}$; the final estimate in (14) is exactly (13) raised to the power $1/\theta$.

Let $U = \sum_k u_k$ and $V = \sum_k v_k$. Discrete Hölder's inequality at each point gives

$$\sum_k f_k \leq U^{1-\theta} V^\theta. \quad (15)$$

The scalar inequality $(\sum a_k)^p \leq \sum a_k^p$ and (14) give

$$\|U\|_p \leq S. \quad (16)$$

Lemma 2 and (14) give

$$\|V\|_{p,\infty} \leq D_p(1-\theta)^{1/(r\theta)} S. \quad (17)$$

Applying Lemma 4 to (15), then using (16) and (17), yields

$$\begin{aligned} \left\| \sum_k f_k \right\|_{p,r} &\leq K_\theta D_p^\theta (1-\theta)^{1/r} S \\ &= \mathcal{H}(\theta)^{1/p} p^{-\theta} (1-p)^{-\theta/p} S. \end{aligned}$$

The last identity uses $1/r = (1 - \theta)/p$ and $D_p = p^{-1}(1 - p)^{-1/p}$. This is (4).

For arbitrary functions, first use $|\sum_k f_k| \leq \sum_k |f_k|$. Approximate each $|f_k|$ increasingly by finite-support simple functions. Such an approximation exists because $\mu(|f_k| > \varepsilon) < \infty$ for every $\varepsilon > 0$. The Fatou property of the Lorentz quasi-norm passes the proved estimate to the limit. Countable families follow by applying the finite result to partial sums and using Fatou once more.

Finally, $\theta/p = 1/p - 1/r$, $\mathcal{H}(\theta) \leq 2$, and

$$p^{-p\theta} \leq p^{-p} \leq e^{1/e} \quad (0 < p < 1).$$

Raising the refined constant in (4) to the p th power therefore gives (5) with $A = 2e^{1/e}$. The logarithm is gone. $\square$

6 Endpoint checks and scope

The refined bound has the correct qualitative interpolation limits. As $r \downarrow p$, one has $\theta \downarrow 0$ and its right-hand constant tends to one, exactly matching $L^{p,p} = L^p$. As $r \rightarrow \infty$, one has $\theta \uparrow 1$ and the constant tends to $D_p = p^{-1}(1 - p)^{-1/p}$, the weak endpoint proved above.

The theorem answers the source's stated yes/no issue: no logarithmic factor is required in a uniform upper bound. It does not claim that the displayed entropy-refined constant is the exact best value of $C(p, r)$. Determining the exact optimum remains a separate extremal problem.

References

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- [2] E. M. Stein, M. H. Taibleson, and G. Weiss, *Weak type estimates for maximal operators on certain H^p classes*, Rend. Circ. Mat. Palermo (2), Suppl. 1 (1981), 81–97.